Dependent n-types characterize n-connected maps

The Univalent Foundations Program

hott-book:578b85cc8d586b1677ec4335148adeb443057d24; CC BY-SA 3.0

Editorial note. Original author statement, proof and supporting context preserved. Only the document wrapper and navigational reference bindings are added; no mathematical repair or correctness certification.

Allowed standard prerequisites (editorial).

- Homotopy type theory. Univalence, dependent products and sums, identity types, propositional and n-truncation; native book references retained.. Standard background, without replacing the counted argument.

Source reference. Original reference 4.4: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1414–1414. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.7.3: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1532–1532. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 8.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1656–1656. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 4.6: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1680–1680. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 7.3.9: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1906–1906. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 7.3.2: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1907–1907. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

7.1 Connectedness

An n-type is one that has no interesting information above dimension n. By contrast, an *n-connected type* is one that has no interesting information *below* dimension n. It turns out to be natural to study a more general notion for functions as well.

Definition 7.1.1. A function $f : A \to B$ is said to be **n-connected** if for all $b : B$, the type $\|\mathsf{fib}_f(b)\|_n$ is contractible:

$$\mathsf{conn}_n(f) :\equiv \prod_{b:B} \mathsf{isContr}(\|\mathsf{fib}_f(b)\|_n).$$

A type A is said to be **n-connected** if the unique function $A \to \mathbf{1}$ is n-connected, i.e. if $\|A\|_n$ is contractible.

Thus, a function $f : A \to B$ is n-connected if and only if $\mathsf{fib}_f(b)$ is n-connected for every $b : B$. Of course, every function is (-2)-connected. At the next level, we have:

Lemma 7.1.2. *A function f is (-1)-connected if and only if it is surjective in the sense of original reference 4.6.*

Proof. We defined f to be surjective if $\|\mathsf{fib}_f(b)\|_{-1}$ is inhabited for all b. But since it is a mere proposition, inhabitation is equivalent to contractibility. □

Thus, n-connectedness of a function for $n \geq 0$ can be thought of as a strong form of surjectivity. Category-theoretically, (-1)-connectedness corresponds to essential surjectivity on objects, while n-connectedness corresponds to essential surjectivity on k-morphisms for $k \leq n + 1$.

Lemma 7.1.2 also implies that a type A is (-1)-connected if and only if it is merely inhabited. When a type is 0-connected we may simply say that it is **connected**, and when it is 1-connected we say it is **simply connected**.

Remark 7.1.3. While our notion of n-connectedness for types agrees with the standard notion in homotopy theory, our notion of n-connectedness for *functions* is off by one from a common indexing in classical homotopy theory. Whereas we say a function f is n-connected if all its fibers are n-connected, some classical homotopy theorists would call such a function $(n + 1)$-connected. (This is due to a historical focus on *cofibers* rather than fibers.)

We now observe a few closure properties of connected maps.

Lemma 7.1.4. *Suppose that g is a retract of a n-connected function f. Then g is n-connected.*

Proof. This is a direct consequence of original reference 4.7.3. □

Corollary 7.1.5. *If g is homotopic to a n-connected function f, then g is n-connected.*

Lemma 7.1.6. *Suppose that $f : A \to B$ is n-connected. Then $g : B \to C$ is n-connected if and only if $g \circ f$ is n-connected.*

Proof. For any $c : C$, we have

$$\begin{aligned}
\|\mathsf{fib}_{g\circ f}(c)\|_n &\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \mathsf{fib}_f(\mathsf{pr}_1 w) \Big\|_n && \text{(by original reference 4.4)} \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \Big\|_n && \text{(by original reference 7.3.9)} \\
&\simeq \|\mathsf{fib}_g(c)\|_n. && \text{(since } \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \text{ is contractible)}
\end{aligned}$$

It follows that $\|\mathsf{fib}_g(c)\|_n$ is contractible if and only if $\|\mathsf{fib}_{g\circ f}(c)\|_n$ is contractible. □

Importantly, n-connected functions can be equivalently characterized as those which satisfy an "induction principle" with respect to n-types. This idea will lead directly into our proof of the Freudenthal suspension theorem in original reference 8.6.

Lemma 7.1.7. *For $f : A \to B$ and $P : B \to \mathcal{U}$, consider the following function:*

$$\lambda s.\, s \circ f : \Big(\prod_{b:B} P(b)\Big) \to \Big(\prod_{a:A} P(f(a))\Big).$$

For a fixed f and $n \geq -2$, the following are equivalent.

(i) f is n-connected.
(ii) For every $P : B \to n\text{-}\mathsf{Type}$, the map $\lambda s.\, s \circ f$ is an equivalence.
(iii) For every $P : B \to n\text{-}\mathsf{Type}$, the map $\lambda s.\, s \circ f$ has a section.

Proof. Suppose that f is n-connected and let $P : B \to n\text{-}\mathsf{Type}$. Then we have the equivalences

$$\begin{aligned}
\prod_{b:B} P(b) &\simeq \prod_{b:B} \Big(\|\mathsf{fib}_f(b)\|_n \to P(b)\Big) && \text{(since } \|\mathsf{fib}_f(b)\|_n \text{ is contractible)} \\
&\simeq \prod_{b:B} \Big(\mathsf{fib}_f(b) \to P(b)\Big) && \text{(since } P(b) \text{ is an } n\text{-type)} \\
&\simeq \prod_{(b:B)} \prod_{(a:A)} \prod_{(p:f(a)=b)} P(b) \\
& && \text{(by the left universal property of } \Sigma\text{-types)} \\
&\simeq \prod_{a:A} P(f(a)). && \text{(by the left universal property of path types)}
\end{aligned}$$

We omit the proof that this equivalence is indeed given by $\lambda s.\, s \circ f$. Thus, (i)⇒(ii), and clearly (ii)⇒(iii). To show (iii)⇒(i), consider the type family

$$P(b) :\equiv \|\mathsf{fib}_f(b)\|_n.$$

Then (iii) yields a map $c : \prod_{(b:B)} \|\mathsf{fib}_f(b)\|_n$ with $c(f(a)) = \big|(a, \mathsf{refl}_{f(a)})\big|_n$. To show that each $\|\mathsf{fib}_f(b)\|_n$ is contractible, we will find a function of type

$$\prod_{(b:B)} \prod_{(w:\|\mathsf{fib}_f(b)\|_n)} w = c(b).$$

By original reference 7.3.2, for this it suffices to find a function of type

$$\prod_{(b:B)} \prod_{(a:A)} \prod_{(p:f(a)=b)} |(a,p)|_n = c(b).$$

But by rearranging variables and path induction, this is equivalent to the type

$$\prod_{a:A} \big|(a, \mathsf{refl}_{f(a)})\big|_n = c(f(a)).$$

This property holds by our choice of $c(f(a))$. □